

New progress on the zeta function: From old conjectures to a major breakthrough

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Early History

The zeta function $\zeta(s)$ today is the oldest and most important tool to study the distribution of prime numbers and is the simplest example of a whole class of similar functions, equally important for understanding the deepest problems of number theory. The celebrated Riemann hypothesis is that all complex zeros of $\zeta(s)$ have real part equal to $\frac{1}{2}$. The consequences of a proof and even of an unlikely disproof of this hypothesis would be a giant step forward for understanding prime numbers.

The paper by Griffin et al. (1) makes fundamental progress in the study of the Riemann zeta function by introducing a method to study certain classical polynomials (the so-called Jensen polynomials) that were known to play a role for understanding the finer properties of the zeta function but had proved to be quite intractable to study by means of standard methods. What was known before this work was a plausible but inaccessible conjecture, called hyperbolicity, for all of them. In this paper the authors introduce a method to study these polynomials which allows the authors to establish hyperbolicity for a very large subset of them.

As an example of the progress made, the random matrix model for the zeta and allied functions was proposed in a fundamental paper by Keating and Snaith (2), putting aside many previous naive attempts to use a Gaussian law to explain the randomness of arithmetical functions. Consider for example tossing coins with the function $\lambda(n)$ which has value +1 or -1 according as the integer n has an even or odd number of prime factors (counting multiplicity). The old prediction of a Gauss law for the deviation from the mean in this random-looking sequence now is considered to be totally wrong and needs to be replaced by a much more sophisticated model, consistent with the random matrix model of Keating and Snaith (2). It is again the Gauss law, but in a subtly defined space of random matrices.

This paper rigorously shows that random matrix theory is needed to explain the oscillations of the classic zeta function. In the geometric setting of arithmetic over fields of characteristic p , this was proved to be the case in a famous work by Katz and Sarnak (3). At last, we have now another big step in the right direction and a further vindication of the insight of mathematical physicists. Undoubtedly, this paper will be of interest to a public going beyond number theorists.

The behavior of $\zeta(s)$ as a function of the complex variable s is very difficult to study. Mathematicians have looked at the Riemann hypothesis from every angle and hundreds of equivalent formulations of it have been made. One of the oldest approaches goes back to Jensen, and his unpublished notes were studied, completed, and expanded by Pólya in detail in 1927 (4).

The conjecture of Jensen is that the roots of the polynomials associated to the Taylor expansion of $s(1-s)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ around the point $s = 1/2$ are all real. This property is called hyperbolicity. If one sets $s = \frac{1}{2} + ix$ one can write the Taylor expansion of this function as $\sum_n c_n x^{2n}/n!$ and, by definition, the sum $\sum_{h=0}^d c_h \binom{d}{h} x^h$ is the Jensen polynomial of degree d . Moreover, one may consider the shifted Jensen polynomials with c_{h+n} in place of c_h . The Jensen criterion for the validity of the Riemann hypothesis is that all of the Jensen polynomials $J^{d,n}(x)$ of degree d and shift n are hyperbolic. In fact, it suffices to prove hyperbolicity just for $n=0$. Older work did this for degree up to 5 and shift 0, and modern computational methods verified it for degrees $d \leq 2 \cdot 10^{17}$ (see ref. 5).

Generalized Jensen polynomials show up in various other contexts, particularly in convex optimization, but only relatively recently have they, and their extension to a theory in several variables, attracted real attention [see, e.g., Bauschke et al. (6)].

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The Main Result

In this relatively short paper, the authors are able to prove the desired hyperbolicity for a big chunk of the original Jensen polynomials, namely for every fixed degree d and all $n \geq N(d)$. Although this remains far away from proving the Riemann hypothesis, it is a big step forward in making progress along Jensen's line of thought. Most importantly, the method can be applied to other situations, and the authors give some interesting examples of this. Their idea is that for large shifts n the coefficients a_{n+j} can be computed

asymptotically with incredible precision. As a consequence, they are able to show that the Riemann zeta function in the derivative aspect Gaussian Unitary Ensemble (GUE) follows the predicted random matrix model GUE after an appropriate renormalization. This is a rigorously proved result of this type and not just conjectured or made plausible by numerical evidence.

There is no doubt that this paper will inspire further fundamental work in other areas of number theory as well as in mathematical physics.

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